SUPPLEMENTARY INFORMATION

Reactive capture of CO_2 via amino acid

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Supplementary Note 1

We compared the total energy expenditure towards CO of amine-based reactive capture to other low-temperature CO_2 electrolysis systems including gas-phase CO_2 electrolysis and hydroxide-based reactive capture. The processes considered in this analysis include CO_2 capture for the initial feedstock, CO_2 electrolysis to produce CO, and CO_2 recovery to separate excess CO_2 from the product gas stream. The values used are based on reported works that demonstrate best-in-class performance. Details are summarized in Supplementary Table 1.

CO_2 Capture

For all systems, it is assumed that 1.57 tonne of CO_2 was initially sourced from the air per 1 tonne of CO produced using a fluid pump for the capture solution (0.08 GJ t_{CO2}^{-1}) and fans for airflow (0.22 GJ t_{CO2}^{-1}) to optimize capture kinetics[1]. For reactive capture systems, CO_2 is regenerated in-situ and therefore only requires 0.47 GJ t_{CO}^{-1} for CO_2 capture. For gas-phase CO_2 electrolysis that requires a pure CO_2 feedstock, additional processes are considered: pellet formation, calcination, slaking, CO_2 dehydration and compression, and air separation for natural gas combustion. These additional steps result in a total of 10.3 GJ t_{CO}^{-1} required for CO_2 capture.

Electrolysis

The CO_2 electrolysis energy was calculated based on the energy efficiency of the system and the Gibbs free energy of CO formation (9.18 GJ t_{CO}^{-1}). For the gas-phase CO_2 system, reports with high CO selectivity and low electrolyzer overpotential were considered[2–4]. Averaging these reports yielded a CO FE of 95.3% and full cell voltage of 3.17 V, corresponding to a CO EE of 40.0% and electrolysis energy of 23.0 GJ t_{CO}^{-1}. For hydroxide-based reactive capture, the values used were from the report with the highest CO FE to date[5]. A CO FE of 46.4% and full cell voltage of 3.79 V were used for energy calculations, which yielded a CO EE of 16.3% and electrolysis energy of 56.4 GJ t_{CO}^{-1}. For this work, the values used were based on the best EE of 31.0%, calculated using a CO FE of 63.8% and full cell voltage of 2.74 V, yielding an electrolysis energy of 29.6 GJ t_{CO}^{-1}.

For this work, operating the electrolyzer at 40 °C does not require additional energy input since the absorption of CO_2 from process gas typically occurs ~40 °C[6–8]. The energy required to pressurize the system was calculated based on batch operation (5 hours of electrolysis), 1 mL of headspace in each of the anolyte and catholyte containers, a constant current density of 50 mA cm^{-2}, and an average 50% CO FE observed during the stability experiment. Over 5 hours, 0.065 g_{CO} was produced. The system was initially pressurized from 1 to 4 bar by Ar, and the pressure was maintained by a downstream pressure relief valve. Therefore, the work for compression was calculated by the following equation, assuming ideal gas behaviour, isothermal compression, and that the fluid is incompressible:

$$W_{\text{comp}}(\text{J}) = \int_{V_1}^{V_2} P\text{d}V = nRT \ln\left|\frac{P_1}{P_2}\right|$$

where n is the number of moles of gas needed to compress the headspace in the catholyte and anolyte containers at the operating temperature (0.31 mmol), R is the ideal gas constant (0.08206 L atm K^{-1} mol^{-1} or 8.314 J K^{-1} mol^{-1}), T is the operating temperature (313 K), P_1 is the initial pressure (1 bar), and P_2 is the final pressure (4 bar). The energy required per batch process was 1.12 J, or 0.02 GJ t_{CO}^{-1}.

CO_2 utilization

CO_2 utilization is measured by the gas composition of the cathode product gas stream and refers to the ratio between converted and total CO_2. CO_2 utilization does not account for in-situ CO_2 that was recaptured by the catholyte and remained in the chemisorbed form. The calculation uses the following equation:

$$\mathrm{CO_2\ utilization\ (\%)} = \frac{n_{\mathrm{CO_2,converted\ to\ CO}}}{n_{\mathrm{CO_2,converted\ to\ CO}} + n_{\mathrm{CO_2,unconverted}}} \times 100\%$$

Where n_{i} is the number of moles of the gas i. Gas-phase CO_2 electrolyzers typically have a 25% CO_2 utilization[9,10]. Hydroxide-based reactive capture has a 100% CO_2 utilization as no CO_2 is detected in the cathode gas stream[5,11]. The amine-based reactive capture system has different CO_2 utilization rates depending on the operating conditions of the system and current density (Supplementary Fig. 27). The CO_2 utilization used for calculation was 8.6%, obtained at the same operating conditions as the best EE experiment. The amount of excess CO_2 in the product gas stream can be calculated based on CO_2 utilization values and the following equation:

$$\mathrm{Excess\ CO_2}(\mathrm{t_{CO_2}\ t_{CO}^{-1}}) = \left(\frac{1}{\mathrm{CO_2\ utilization}} \times 100\% - 1\right) \times \frac{M_{\mathrm{CO_2}}}{M_{\mathrm{CO}}}$$

Where M_i is the molar mass of the gas i. This yields excess CO_2 of 4.7 t_{CO2} t_{CO}^{-1} and 16.7 t_{CO2} t_{CO}^{-1} for gas-phase CO_2 electrolyzer and amine-based reactive capture, respectively.

CO_2 recovery

Amine absorption is conventionally used to separate concentrated CO_2 from a product stream. The typical energy breakdown for amine-based CO_2 capture is 60% reboiler (4.0 GJ t_{CO2}^{-1}), 30% compressor (2.0 GJ t_{CO2}^{-1}), 8% blower (0.5 GJ t_{CO2}^{-1}), and 2% pumps (0.1 GJ t_{CO2}^{-1})[12]. We assumed that the CO_2 will be separated on-site and fed back into the electrolyzer, eliminating the need for the compressor. Therefore, the energy required to recover CO_2 for the gas-phase system is 4.6 GJ t_{CO2}^{-1}. For the reactive capture systems, a reboiler was not required because the post-capture solution can be used directly in the electrolyzer, yielding a CO_2 recovery energy of 0.6 GJ t_{CO2}^{-1}. Based on the CO_2 utilization, we obtained 21.7 GJ t_{CO}^{-1} and 10.0 GJ t_{CO}^{-1} for gas-phase CO_2 electrolysis and amine-based reactive capture, respectively. We assumed that the gas product will be used in a syngas mixture downstream; therefore, the separation of CO and H_2 was omitted for all systems.

Supplementary Table 1 | Calculation details for the comparison of energy towards CO for gas-phase CO_2 electrolysis, hydroxide- and amine-based reactive capture

		Gas-phase CO_2 electrolysis	Hydroxide-based reactive capture	Amine-based reactive capture (*This work*)
CO_2 capture (GJ t_{CO}^{-1})	Fluid pump (GJ t_{CO2}^{-1})	0.08	0.08	0.08
	Fan (GJ t_{CO2}^{-1})	0.22	0.22	0.22
	Additional CO_2 capture processes (GJ t_{CO2}^{-1})	6.27		
	CO_2 capture total (GJ t_{CO2}^{-1})	6.57	0.30	0.30
	CO_2 capture total (GJ t_{CO}^{-1})	**10.3**	**0.47**	**0.47**
CO_2 electrolysis (GJ t_{CO}^{-1})	CO FE (%)	95.3	46.4	63.8
	Full cell voltage (V)	3.17	3.79	2.74
	Thermodynamic minimum voltage (V)	1.33	1.33	1.33
	Energy efficiency (%)	40.0	16.3	31.0
	Gibbs free energy of CO (GJ t_{CO}^{-1})	9.18	9.18	9.18
	System pressurization (GJ t_{CO}^{-1})			0.02
	CO_2 electrolysis total (GJ t_{CO}^{-1})	**23.0**	**56.4**	**29.7**
CO_2 utilization (%)	CO_2 utilization (%)	25	100	8.6
	Excess CO_2 (t_{CO2} t_{CO}^{-1})	4.7	0.0	16.7
CO_2 recovery (GJ t_{CO}^{-1})	Pump (GJ t_{CO2}^{-1})	0.1		0.1
	Blower (GJ t_{CO2}^{-1})	0.5		0.5
	Reboiler (GJ t_{CO2}^{-1})	4.0		
	CO_2 recovery total (GJ t_{CO2}^{-1})	4.6	0	0.6
	CO_2 recovery total (GJ t_{CO}^{-1})	**21.7**	**0.0**	**10.0**
Total (GJ t_{CO}^{-1})		**55.0**	**56.8**	**40.2**

Supplementary Table 2 | Comparison of amino acids for CO_2 capture. CO_2 loading capacity was obtained at CO_2 partial pressures ≤ 15 kPa. Cost was obtained from MilliporeSigma for purities ≥ 98% and at the lowest unit price.

Amino acid	CO_2 absorption flux ($mol\ m^2\ h^{-1}$)[13]	CO_2 loading capacity ($mol_{CO2}\ mol^{-1}$)[14]	Cost ($\$\ kg^{-1}$)
Glycine	0.5	0.7	43
Lysine	0.45	1.3	1164
Proline	0.55	0.6	154
Serine	0.35	0.6	613
Threonine	0.35	0.6	712
Arginine	0.65	N/R	67
Tyrosine	0.5	N/R	266
Cystine	0.5	N/R	242
Taurine	N/R	0.5	44

Supplementary Table 3 | Measured conductivities of different chemisorbed forms of CO_2

Electrolyte	Conductivity ($mS\ cm^{-1}$)
2 M $KHCO_3$	122
1 M K_2CO_3	126
2 M K-GLY (after purging)	85

Supplementary Table 4 | Measured weight and concentration of captured CO_2. The post-capture solution has a volume of 75 mL and pH of 8.1. Error bars represent the standard deviation of at least three independent measurements.

Capture solution	Weight of captured CO_2 (g)	Concentration of captured CO_2 (M)
1 M K-GLY + 0.1 M KH_2PO_4	1.96 ± 0.03	0.59 ± 0.01
2 M K-GLY + 0.1 M KH_2PO_4	3.6 ± 0.3	1.08 ± 0.08
3 M K-GLY + 0.1 M KH_2PO_4	5.2 ± 0.3	1.6 ± 0.1

Supplementary Table 5 | Measured CO_2 in the cathode outlet at different temperatures. Error bars represent the standard deviation of at least three independent measurements.

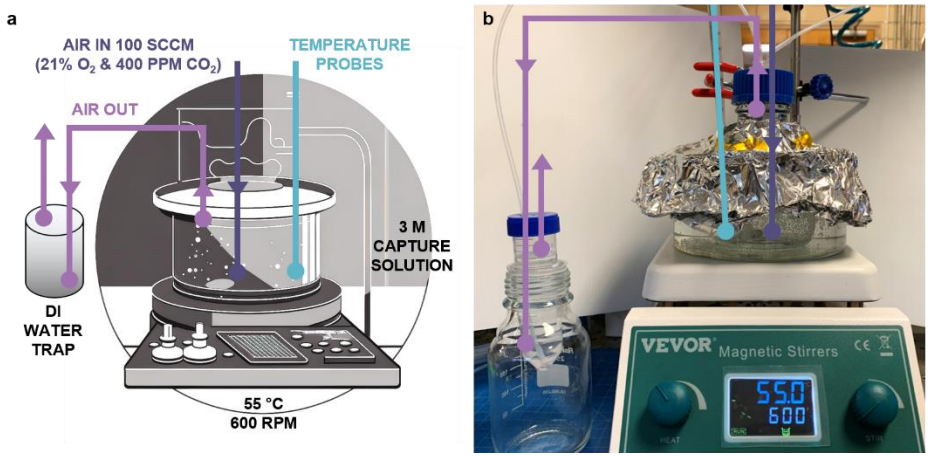
Temperature (°C)	CO_2 (mL s^{-1})
20	0.018 ± 0.001
35	0.035 ± 0.007
40	0.06 ± 0.01
45	0.10 ± 0.03

Supplementary Table 6 | Comparison of the electrochemical performance with amine- and hydroxide-based reactive CO_2 capture reports

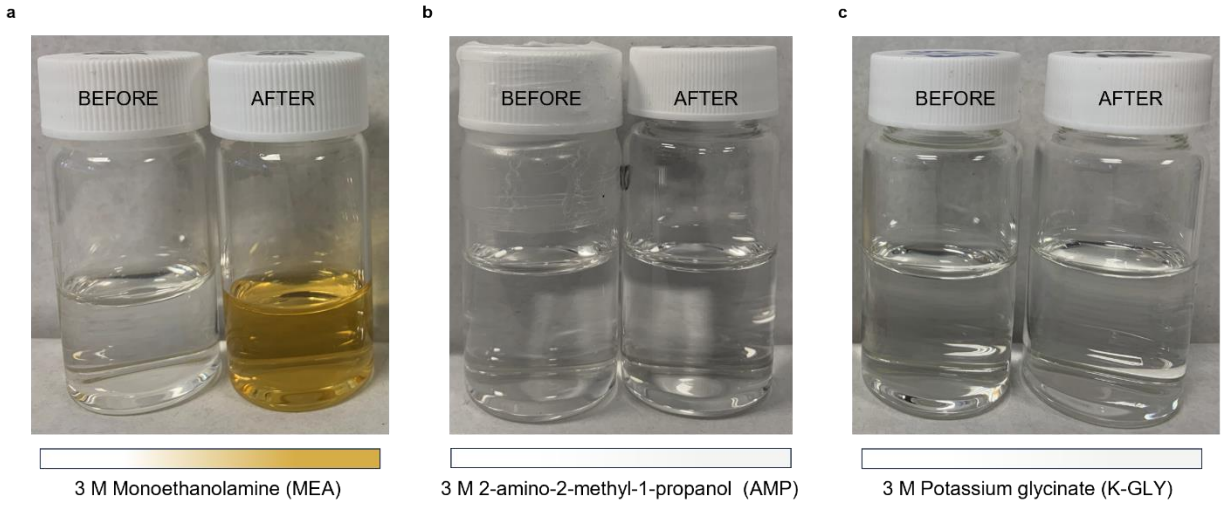
Capture solution	CO FE (%)	Current density (mA cm^{-2})	j_{CO} (mA cm^{-2})	Cathode potential (-V vs. RHE)	Full cell potential (V)	CO EE (%)	Ref.
Amine-based reactive capture							
2 M K-GLY + 0.1 M KH_2PO_4	63.8	50	32		2.74	**31**	*This work*
	44.4	200	**89**		3.17	19	
3 M triethylamine (TREA)	70	20	14				15
	37	100	37		3.5	14	
	30	200	60				
5 M MEA	78.3	2.6	2	0.63			16
	64.9	50	32				
5 M MEA + 2 M KCl	72	50	36	0.8			17
0.4 M methyldiethanolamine (MDEA)-based deep eutectic solvent (DES)	71	20	14	1.1			18
1 M AMP in polycarbonate (PC)	45	15	7	1.6			19
Ethylenediamine (EDA)	45	18.4	8	0.78			20
Hydroxide-based reactive capture							
2 M KOH	46	200	93		3.8	16	5
2 M KOH	28	180	50		3.8	10	11

Supplementary Table 7 | Flue gas composition from various sources. Values are obtained from the literature and industry[21–27]

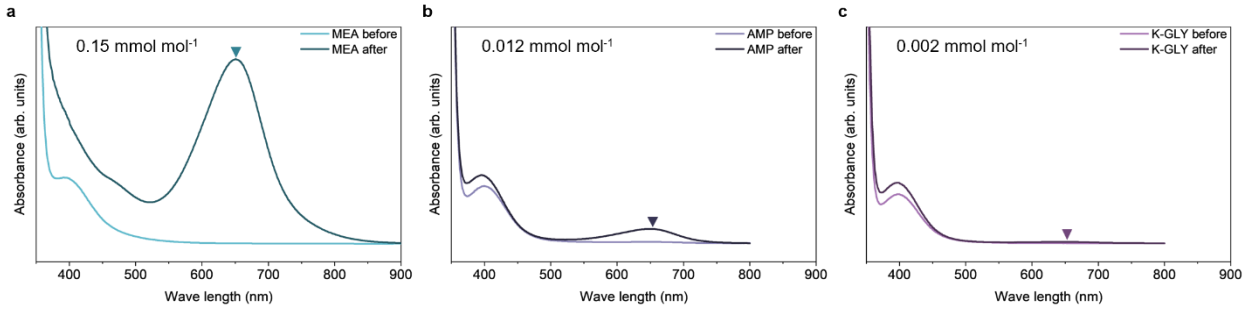
Component	Natural gas combined cycle (NGCC)	Coal-fired heater	Cement	Iron and steel	Ethanol	Steam-methane reforming (SMR)
CO_2	4 – 10%	10 – 12%	14 – 33%	1 – 30%	97%	20%
N_2	67 – 75%	73 – 76%	56 – 68%	20 – 55%	2%	50%
O_2	5 – 12%	3 – 5%	2 – 14%	0 – 1%	1%	1%
H_2O	7 – 15%	7 – 8%	7 – 18%	0 – 5%	0%	18%



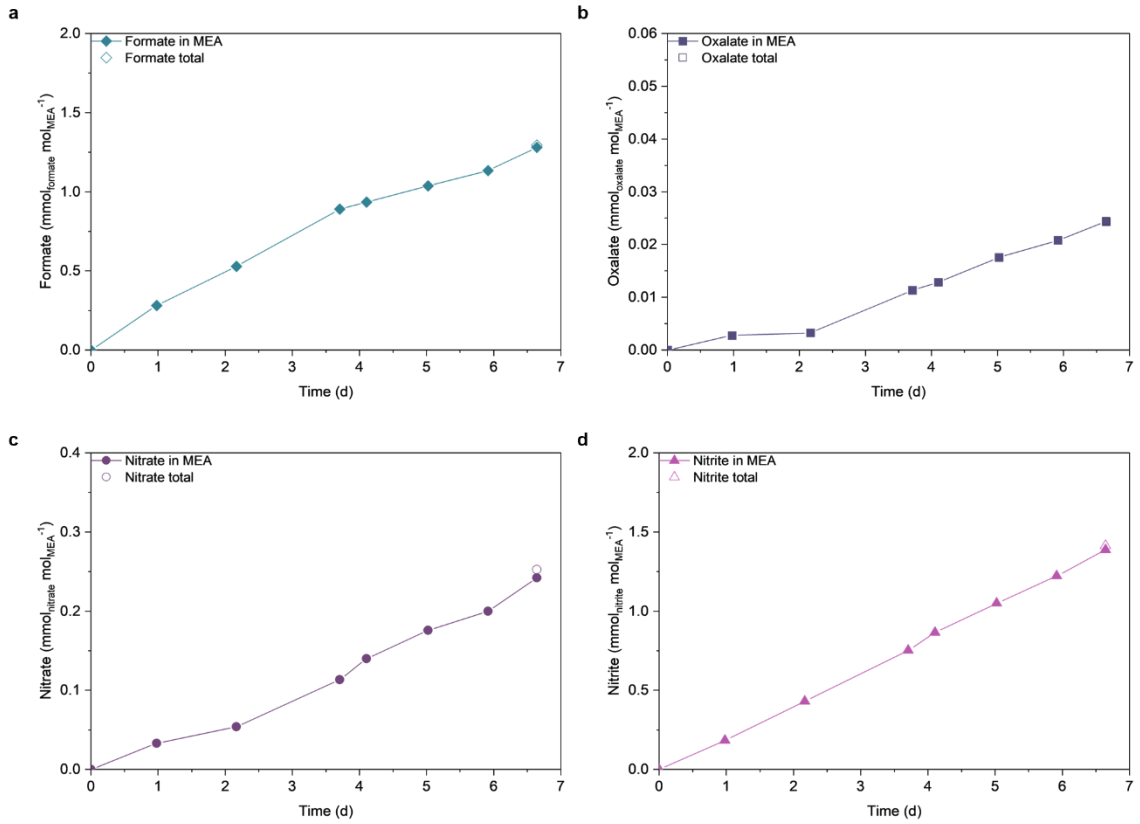
Supplementary Fig. 1 | Experimental setup for the expedited static oxygen degradation of capture solutions. a, schematic and **b**, photograph of the setup.



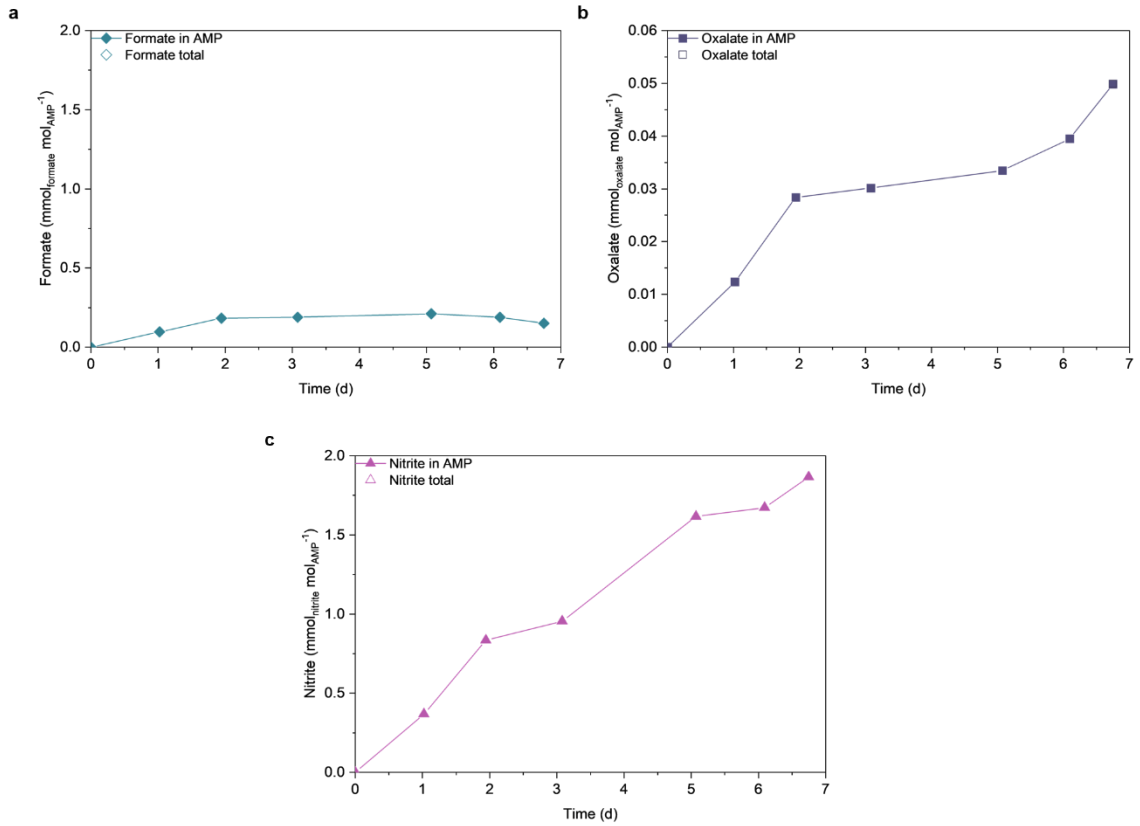
Supplementary Fig. 2 | Photographs of the capture solutions before and after the 7-day expedited static oxygen degradation experiment. a, 3 M MEA capture solution. **b**, 3 M AMP capture solution. **c**, 3 M K-GLY capture solution.



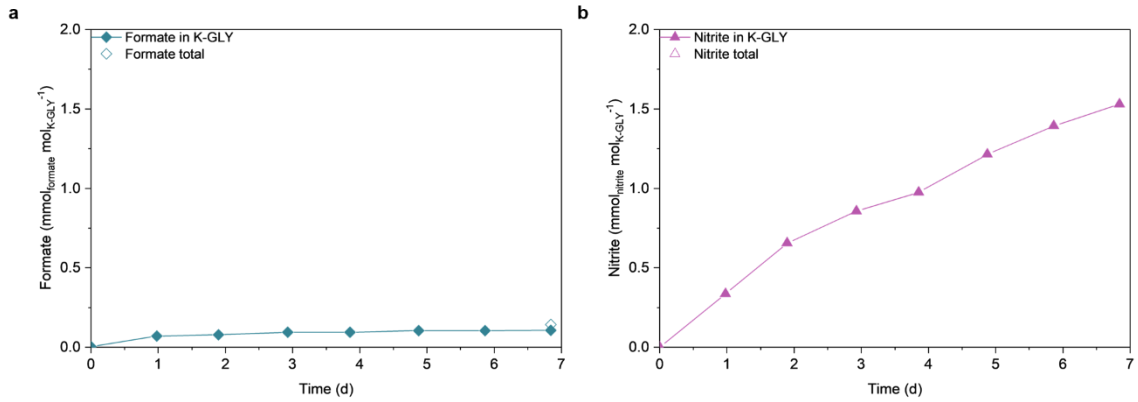
Supplementary Fig. 3 | Ultraviolet-visible spectroscopy of the initial capture solution and of the DI water trap liquid after the 7-day expedited static oxygen degradation experiment. a, 3 M MEA capture solution. **b**, 3 M AMP capture solution. **c**, 3 M K-GLY capture solution. Ammonia was quantified by the absorbance at a wavelength of 655 nm, labelled by an upside-down triangle marker (▼).



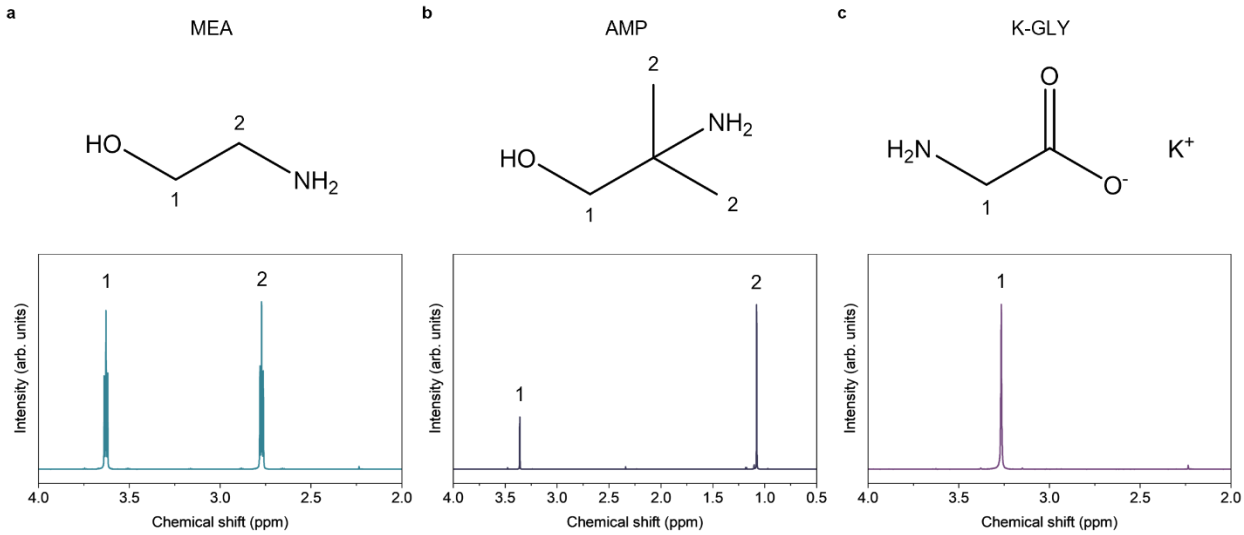
Supplementary Fig. 4 | Concentrations of products during the expedited static oxygen degradation of MEA. a, formate. **b**, oxalate. **c**, nitrate. **d**, nitrite. Liquid samples were collected daily from the capture solution and at the end of the experiment from the DI water trap. Total product is the sum of product in the capture solution and product in the DI water trap. No significant amounts of acetate and glycolate were detected. Source data are provided as a Source Data file.



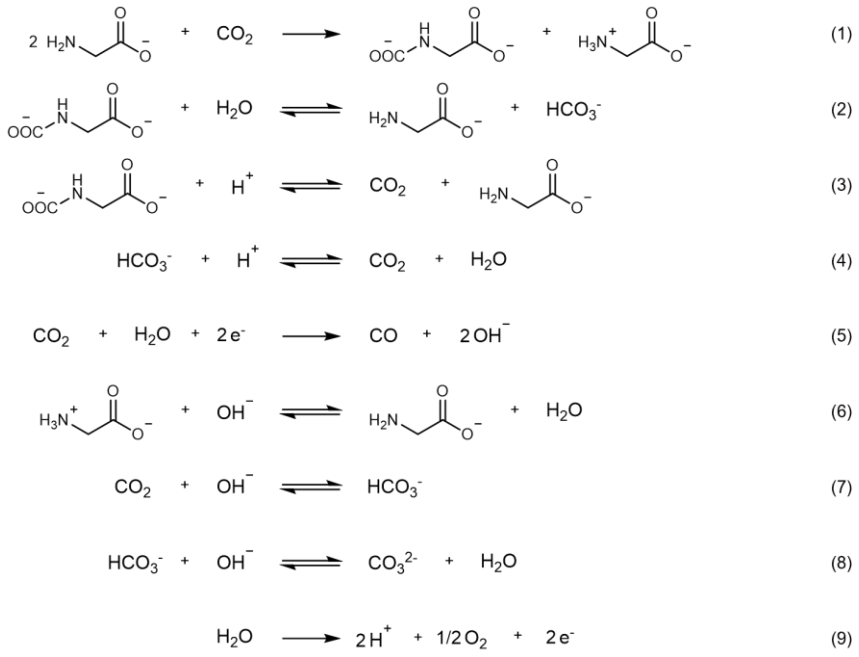
Supplementary Fig. 5 | Concentrations of products during the expedited static oxygen degradation of AMP. a, formate. **b**, oxalate. **c**, nitrite. Liquid samples were collected daily from the capture solution and at the end of the experiment from the DI water trap. Total product is the sum of product in the capture solution and product in the DI water trap. No significant amounts of nitrate, acetate, and glycolate were detected. Source data are provided as a Source Data file.



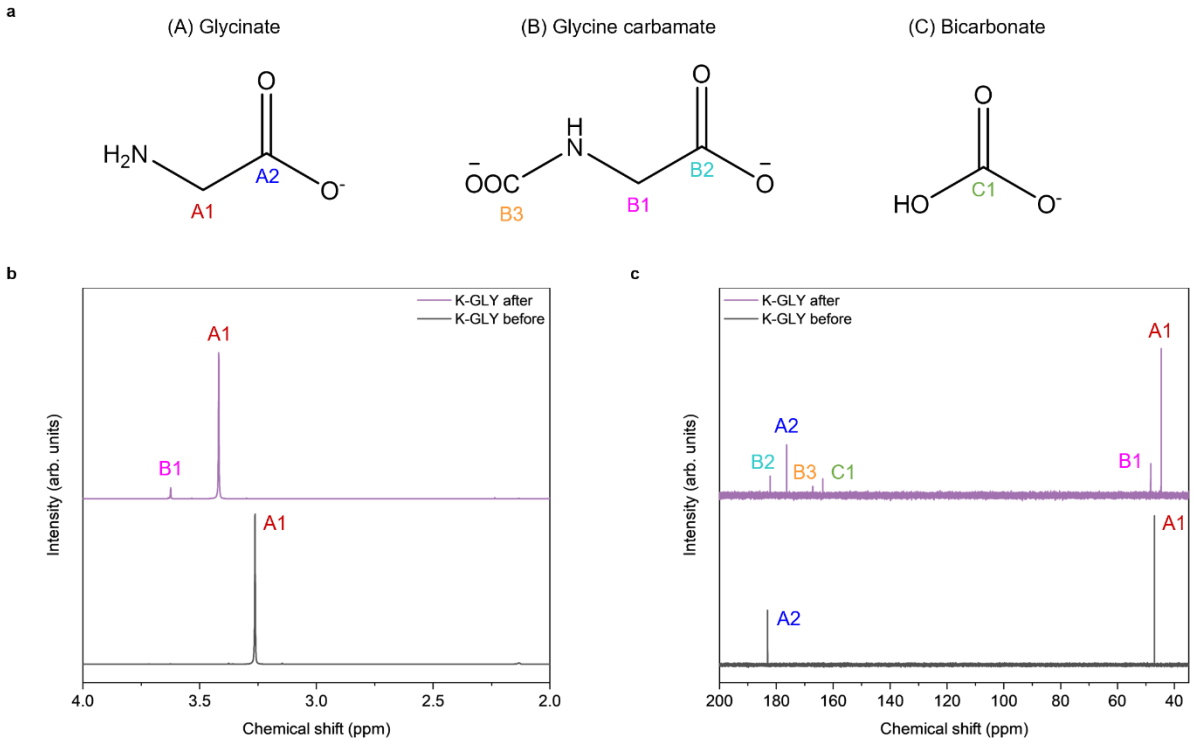
Supplementary Fig. 6 | Concentrations of products during the expedited static oxygen degradation of K-GLY. a, formate. **b**, nitrite. Liquid samples were collected daily from the capture solution and at the end of the experiment from the DI water trap. Total product is the sum of product in the capture solution and product in the DI water trap. No significant amounts of oxalate, nitrate, acetate, and glycolate were detected. Source data are provided as a Source Data file.



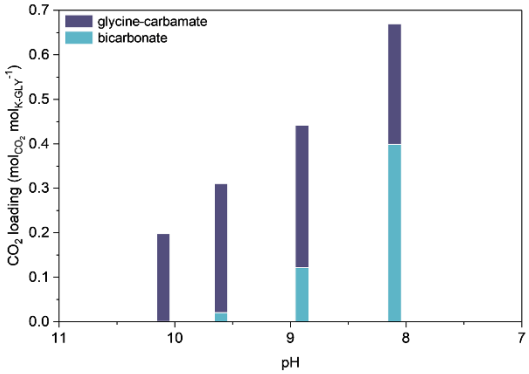
Supplementary Fig. 7 | Chemical structures and ^{1}H NMR spectra of the capture solutions. a, MEA. **b**, AMP. **c**, K-GLY. All samples were diluted with deuterium oxide (D_2O) and used trimethylsilylpropanoic acid (TSP) as the internal standard.



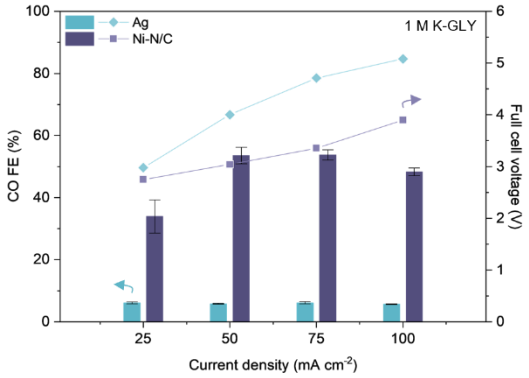
Supplementary Fig. 8 | Chemical reactions involved in the reactive capture system using K-GLY as the capture solution.



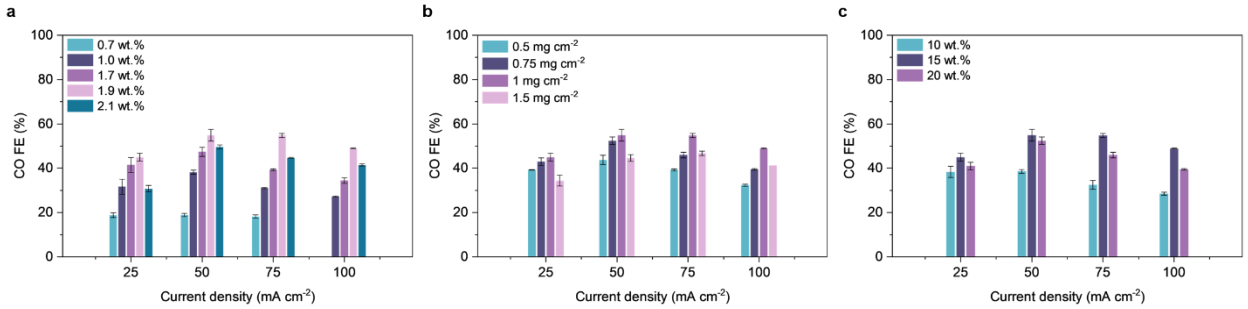
Supplementary Fig. 9 | Chemical structures of species before and after K-GLY CO_2 capture. **a,** chemical structures of the glycinate ion, the glycine carbamate ion, and the bicarbonate ion. **b**, ^{1}H NMR spectra with peaks of the glycinate ion/glycine zwitterion and glycine carbamate ion labelled. **c**, ^{13}C NMR spectra with peaks of the glycinate ion/glycine zwitterion, glycine carbamate ion, and bicarbonate ion labelled. All samples were diluted with D_2O and used trimethylsilylpropanoic acid (TSP) as the internal standard. Peak positions were confirmed with previously published results[28–30].



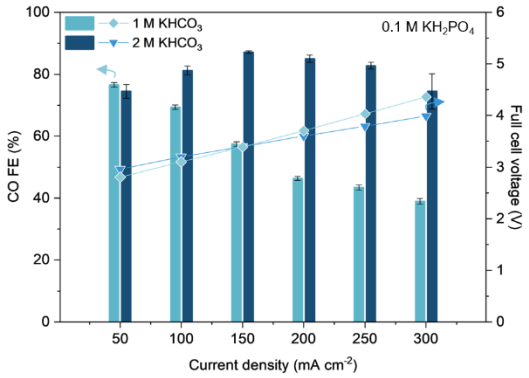
Supplementary Fig. 10 | Chemical composition of chemisorbed CO_2 in the post-capture solution at various pH. Source data are provided as a Source Data file.



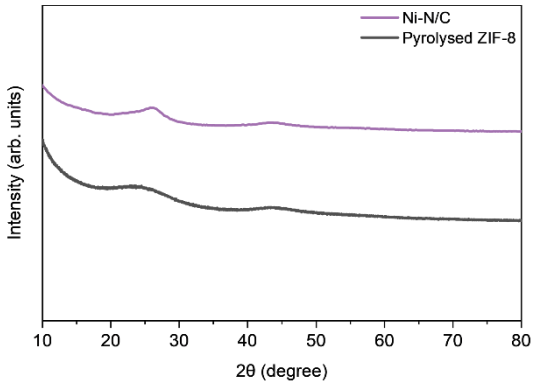
Supplementary Fig. 11 | Electrolysis performance with an Ag catalyst and a Ni-N/C catalyst. FE towards CO and full cell voltage using a 1 M K-GLY capture solution. Error bars represent the standard deviation of at least three independent measurements. Source data are provided as a Source Data file.



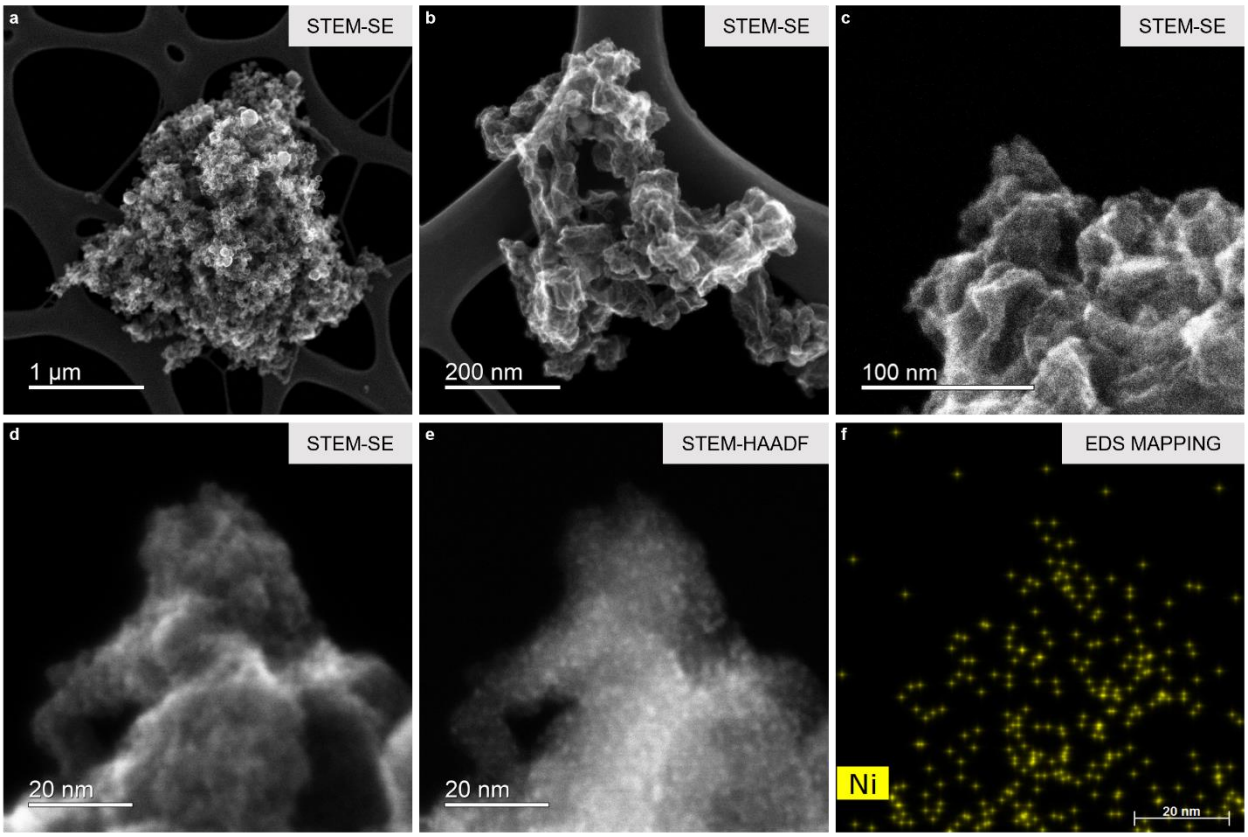
Supplementary Fig. 12 | Electrolysis performance for various catalyst compositions. a, FE towards CO using a Ni-N/C catalyst with nickel weight percent varying from 0.7 wt.% to 2.1 wt.%. **b**, FE towards CO using a Ni-N/C catalyst with catalyst loading onto the current conducting substrate varying from 0.5 mg cm^{-2} to 1.5 mg cm^{-2}. **c**, FE towards CO using a Ni-N/C catalyst with Nafion ionomer weight percent varying from 10 wt.% to 20 wt.%. Error bars represent the standard deviation of at least three independent measurements. Source data are provided as a Source Data file.



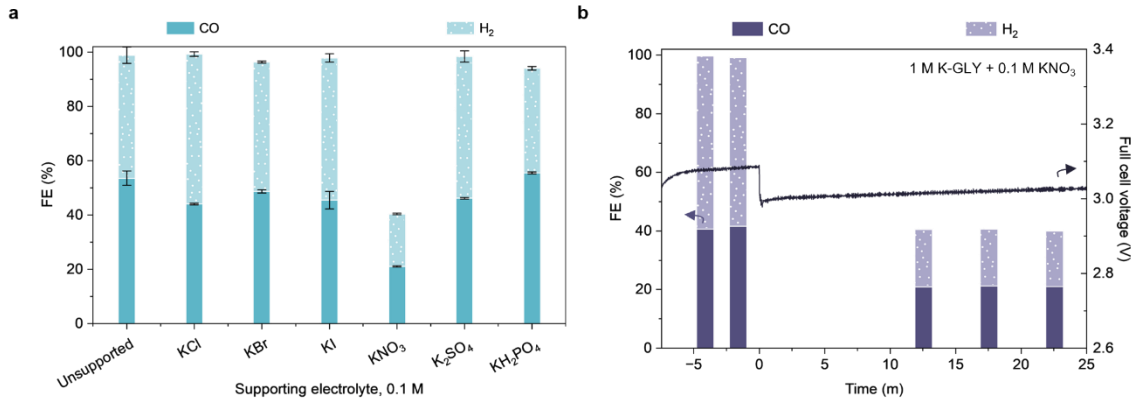
Supplementary Fig. 13 | Electrolysis performance using pure potassium bicarbonate ($KHCO_3$) as the carbon source. FE towards CO and full cell voltage using a 1 M and 2 M $KHCO_3$ capture solution with 0.1 M KH_2PO_4. Error bars represent the standard deviation of at least three independent measurements. Source data are provided as a Source Data file.



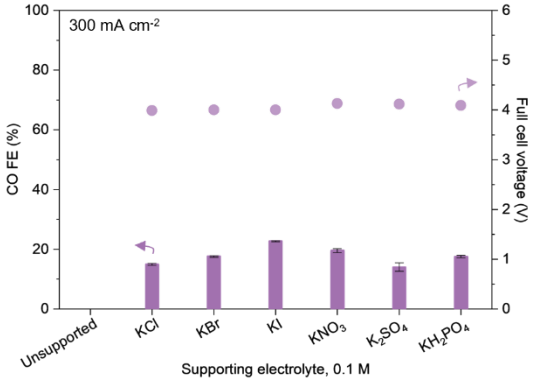
Supplementary Fig. 14 | XRD spectra of Ni-N/C and pyrolysed ZIF-8. No Ni or NiO characteristic peaks were observed for the Ni-N/C powder sample[31]. Two broad peaks present in both powder samples correspond to carbon planes[32,33].



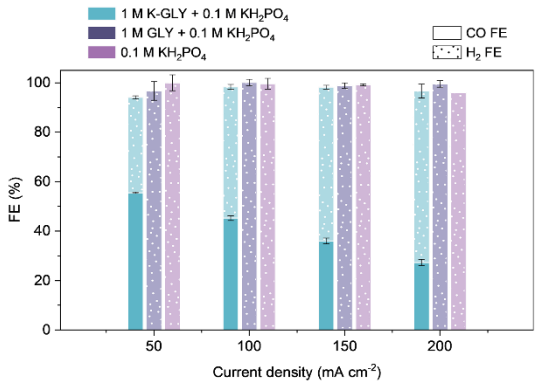
Supplementary Fig. 15 | Scanning transmission electron microscopy (STEM) images of the Ni-N/C catalyst. a – **d**, STEM images in secondary electron (SE) imaging mode. **e**, STEM image in high angle annular dark-field (HAADF) imaging mode. **f**, energy dispersive X-ray spectroscopy (EDS) mapping of Ni.



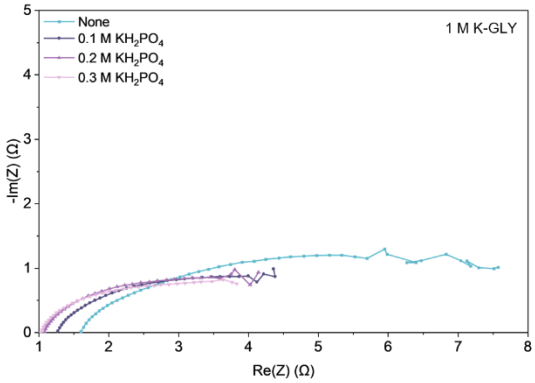
Supplementary Fig. 16 | Total CO and H_2 FE with various potassium salts as supporting electrolyte. a, FE towards CO and H_2 at an applied current density of 50 mA cm^{-2}. Capture solution contains 1 M K-GLY and 0.1 M of supporting electrolyte. When KNO_3 was employed as the supporting electrolyte, the selectivity towards CO and H_2 both decreased due to the thermodynamically more favorable nitrate reduction reaction[24,34,35]. **b**, total FE and full cell voltage over time at 50 mA cm^{-2} when 0.1 M KNO_3 was added to a 2 M K-GLY post-capture solution at time = 0 minutes. Error bars represent the standard deviation of at least three independent measurements. Source data are provided as a Source Data file.



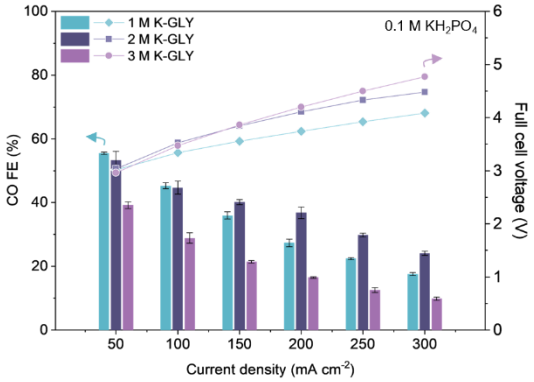
Supplementary Fig. 17 | Electrolysis performance with various potassium salts as supporting electrolyte. FE towards CO and full cell voltage at an applied current density of 300 mA cm^{-2}. Capture solution contains 1 M K-GLY and 0.1 M of supporting electrolyte. Error bars represent the standard deviation of at least three independent measurements. Source data are provided as a Source Data file.



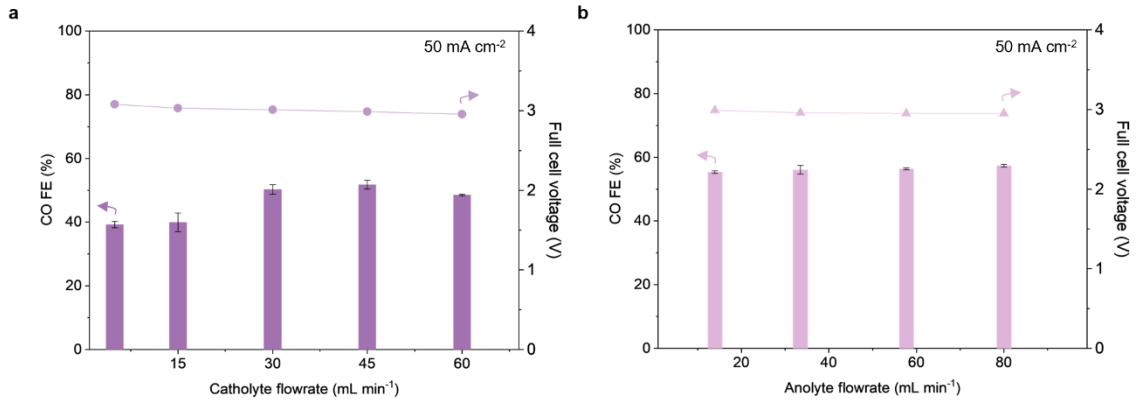
Supplementary Fig. 18 | Electrolysis performance with and without the capture agent K-GLY. FE towards CO and H_2 using various catholytes. Error bars represent the standard deviation of at least three independent measurements. Source data are provided as a Source Data file.



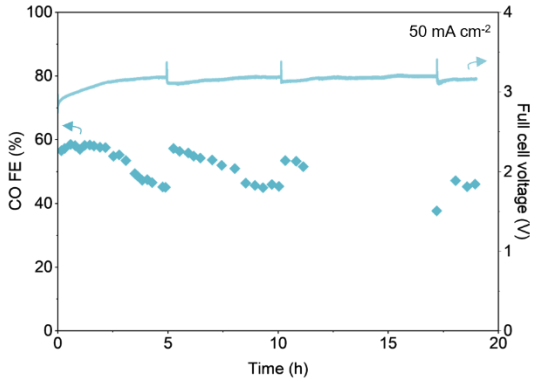
Supplementary Fig. 19 | Electrochemical impedance spectroscopy with various concentrations of KH_2PO_4 supporting electrolyte. Full cell impedance spectra at 75 mA cm^{-2} for a 1 M K-GLY capture solution with KH_2PO_4 supporting electrolyte concentration varying from 0 to 0.3 M. Source data are provided as a Source Data file.



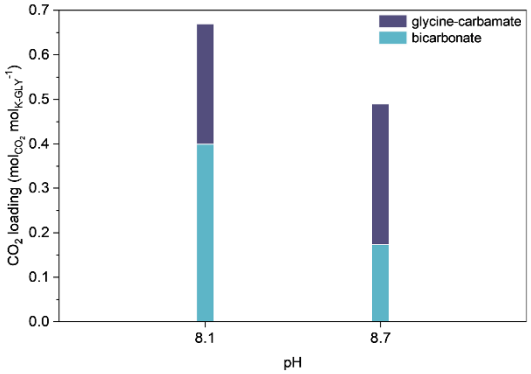
Supplementary Fig. 20 | Electrolysis performance with various concentrations of capture solution. FE towards CO and full cell voltage for K-GLY capture solutions varying from 1 to 3 M supported by 0.1 M KH_2PO_4. Error bars represent the standard deviation of at least three independent measurements. Source data are provided as a Source Data file.



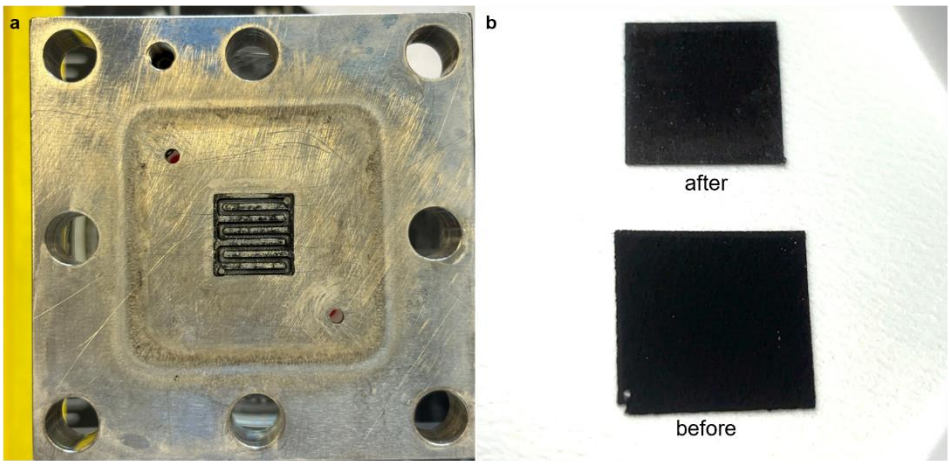
Supplementary Fig. 21 | Electrolysis performance with various electrolyte flowrates. a, FE towards CO and full cell voltage for catholyte flowrates varying from 5 to 60 mL min^{-1}. The anolyte flowrate was kept constant at 35 mL min^{-1}. **b**, FE towards CO and full cell voltage for anolyte flowrates varying from 14 to 80 mL min^{-1}. The catholyte flowrate was kept constant at 45 mL min^{-1}. Error bars represent the standard deviation of at least three independent measurements. Source data are provided as a Source Data file.



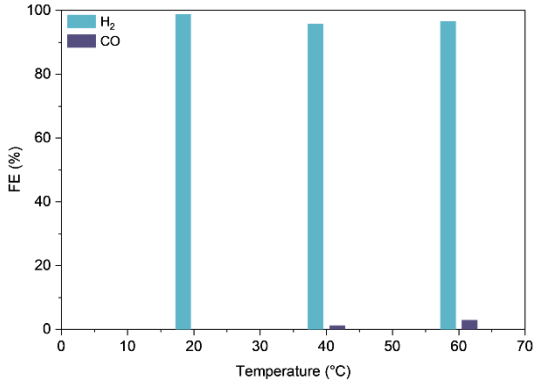
Supplementary Fig. 22 | Stability test of reactive CO_2 capture for 19 hours at an applied current density of 50 mA cm^{-2}. 75mL of catholyte and anolyte were replaced with fresh electrolytes at 5-, 10-, and 17- hour marks. The CO FE increased each time after electrolyte refresh. Source data are provided as a Source Data file.



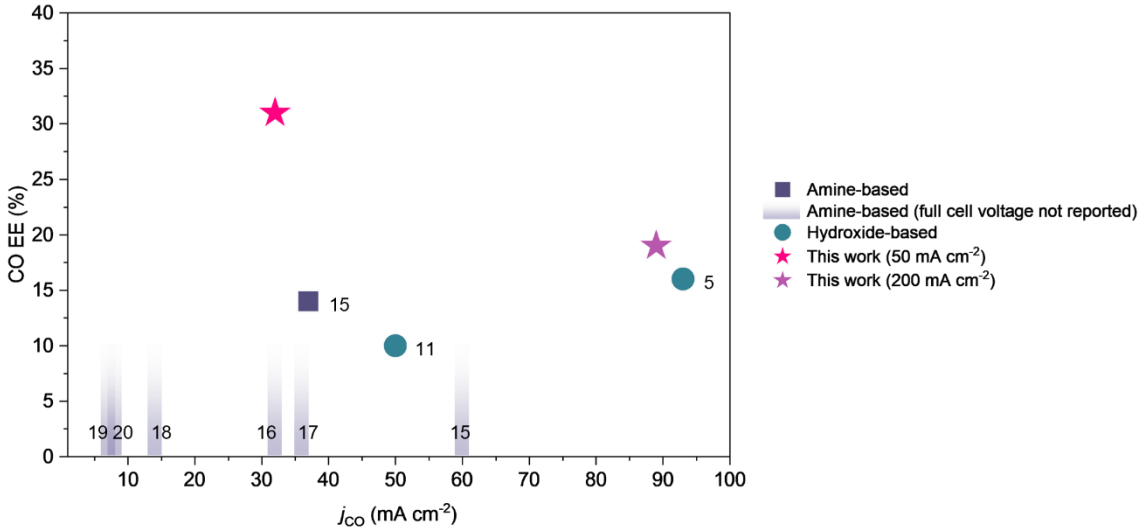
Supplementary Fig. 23 | Chemical composition of chemisorbed CO_2 in the post-capture solution before and after 5 hours of electrolysis at 50 mA cm^{-2}. Source data are provided as a Source Data file.



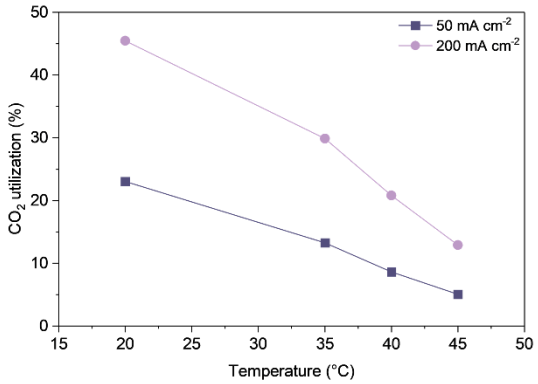
Supplementary Fig. 24 | Photographs of the electrolyzer and catalyst after 32 hours of electrolysis. a, the electrolyzer with catalyst residue on the flow channel. **b**, comparison of the catalyst before and after electrolysis. This observation suggests that mechanical stability of the catalyst could be one possible failure mechanism. Increasing the binding energy between the metal single atom and the support and improving the adherence of the catalyst onto the current conducting substrate may improve the catalyst stability[36].



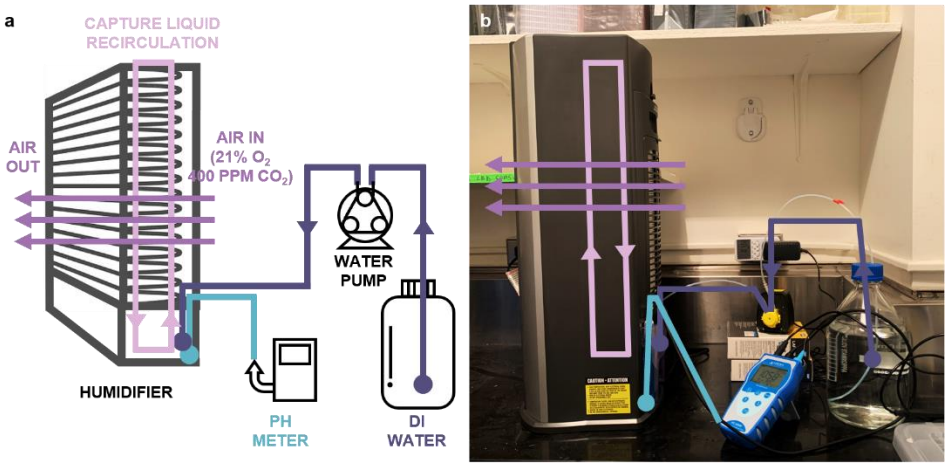
Supplementary Fig. 25 | Electrolysis performance using 1 M potassium hydroxide (KOH) as the anolyte. The capture solution was 1 M K-GLY with 0.1 M KH_2PO_4. The electrolyzer was operated at an applied current density of 50 mA cm^{-2} and temperatures of 20, 40, and 60 °C. Source data are provided as a Source Data file.



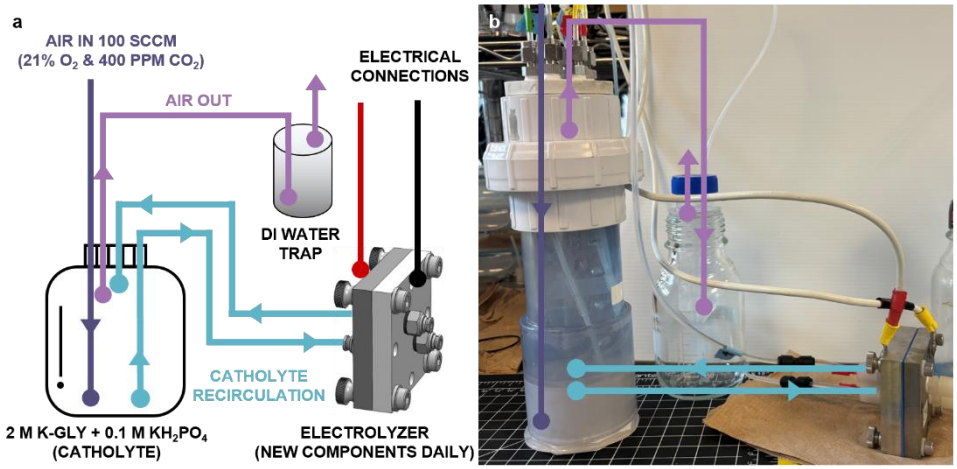
Supplementary Fig. 26 | Summary of state-of-art performance of amine- and hydroxide-based reactive capture systems. For works that did not report the full cell voltage, the CO EE was not calculated and only the j_{CO} is shown in light purple. Details are tabulated in Supplementary Table 6.



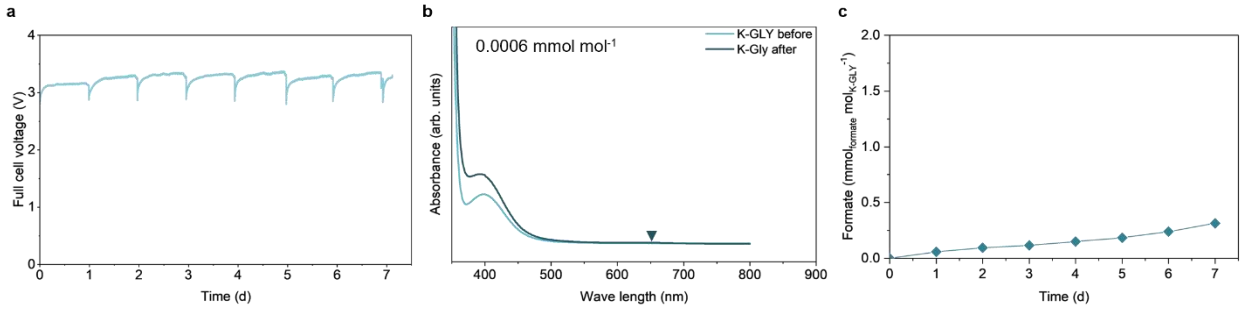
Supplementary Fig. 27 | CO_2 utilization at different temperatures and current densities. The capture solution was 2 M K-GLY with 0.1 M KH_2PO_4. The electrolyzer was operated at a pressure of 4 bar. Source data are provided as a Source Data file.



Supplementary Fig. 28 | Experimental setup for direct air capture (DAC). a, schematic and **b**, photograph of the setup. A household humidifier was used as the air contactor[30,37].



Supplementary Fig. 29 | Experimental setup for oxygen degradation of the capture solution under electrolysis. a, schematic and **b**, photograph of the setup. The electrolyzer assembly was replaced daily.



Supplementary Fig. 30 | Analysis for oxygen degradation of the capture solution under electrolysis. a, full cell voltage of the reactive capture electrolyzer operated at 50 mA cm^{-2}. The spike in full cell voltage each day corresponds to when the electrolyzer assembly was replaced. **b**, ultraviolet-visible spectroscopy of the initial capture solution and of the DI water trap liquid after 7 days. Ammonia was quantified by the absorbance at a wavelength of 655 nm, labelled by an upside-down triangle marker (▼). **c**, Concentration of formate in the capture solution. Liquid samples were collected daily from the capture solution and at the end of the experiment from the DI water trap. No significant amounts of oxalate, nitrate, nitrite acetate, and glycolate were detected. The formate concentration for degradation under electrolysis were higher than that of the expedited static degradation likely due to CO_2 electrolysis into formate[38]. The FE towards formate was < 0.1%. Source data are provided as a Source Data file.

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